



# Update profile

Inverse design: from an EI and Cross Section model and a target EI curve, propose the thickness profile that hits the target.



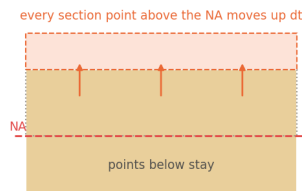
# What it does

Knowing the EI you want is easy; knowing the core thickness that gives it is not. Update profile uses the modelled sections to solve, station by station, for the thickness that meets a target EI curve.

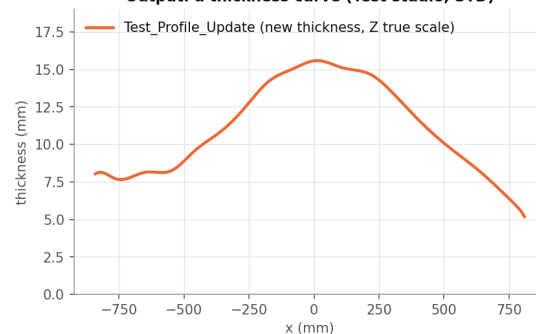
## Use it when

- Hit a target flex from an existing layup
- Scale a design part-way toward a target
- Estimate thickness changes quickly (DELTA / PERCENT)

**STD: find  $dt$  so  $EI(dt) = \text{target}$  (bisection)**



**Output: a thickness curve (Test studio, STD)**

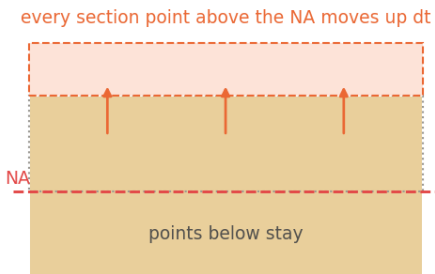


*STD: section points above the neutral axis move up; output is a thickness curve.*

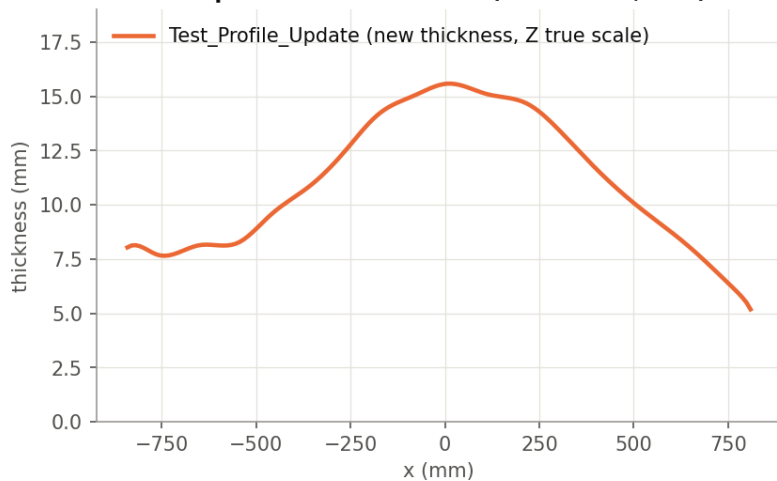


# Three solvers

**STD: find  $dt$  so  $EI(dt) = \text{target}$  (bisection)**



**Output: a thickness curve (Test studio, STD)**



## STD

Full section model: move every point above the NA up by  $dt$ ; bisection on  $dt$  until  $EI = \text{target}$ .

## DELTA

Fit  $EI / \text{width} = \alpha t^\beta$  over all stations; solve  $t$  for the new  $EI$ .

## PERCENT

Scale thickness by the  $EI$  ratio, using the fitted  $\beta$ .



## The dialog (STD) (1/2)

Update profile 1 ✓ ✗

1 **STD** DELTA PERCENT

2 Output curve name **Test\_Profile**

3 Cross section feature  
EI and Cross Section 1 ✕

4 Target EI profile (Q)  
Edge of Measured\_EI ✕

5 Measured EI from QUERY ▼  
Measured EI edges (Q)  
Edge of EI and Cross Sect... ✕

6 Measured thickness from  
INHERIT ▼

| # | Parameter               | What it means                                                                                                                                   |
|---|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Solver type             | STD / DELTA / PERCENT.                                                                                                                          |
| 2 | Output curve name       | Default 'Updated thickness'.                                                                                                                    |
| 3 | Cross section feature   | The EI and Cross Section feature.                                                                                                               |
| 4 | Target EI profile       | The EI curve: an edge in the XZ plane, world Z in mm = EI in N m <sup>2</sup> (EI and Cross Section's <name>_EI, a measured or a target curve). |
| 5 | Measured EI from        | Inherit (the model) or Query (edges).                                                                                                           |
| 6 | Measured thickness from | Inherit or Query (the Query edges are not used).                                                                                                |



## The dialog (STD) (2/2)

7 ☒ Scale delta

Delta scale factor

8 FCP  (Q)

Vertex of Derived 1

9 ACP  (Q)

Vertex of Derived 1

> Stiffness estimates

10 ☒ Spline Output

☐ FIT ☒ APPROXIMATE

Approximation tolerance

Maximum control points

Curve degree

11

| #  | Parameter     | What it means                                                           |
|----|---------------|-------------------------------------------------------------------------|
| 7  | Scale delta   | Apply only part of the change: measured + factor x (target - measured). |
| 8  | FCP           | With ACP: read-only flex numbers of the target.                         |
| 9  | ACP           | Aft contact point.                                                      |
| 10 | Spline Output | Fit / Approximate.                                                      |
| 11 | Recalculate   | No effect today.                                                        |



# Outputs and messages

## Outputs

| Key                                  | What it holds                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| <Output curve name>                  | Thickness wire, Z = thickness, true scale.                                                             |
| <b>profileUpdates</b><br>(attribute) | Per station: x, measured and updated thickness and EI; shown by the Profile Updates table (+ dt, dEI). |

## Messages

|        | Message / what to do                                                |
|--------|---------------------------------------------------------------------|
| (none) | A spline fit that fails draws red / magenta debug points; no error. |



## Tips and limits



### Growth above the NA

STD assumes added thickness goes on top (core / topsheet); a base-side change is not modelled.



### Coarse target read

The target EI is sampled 11 times per edge; stations outside it are skipped.



### Guard

A new thickness outside 0.1-200 mm is reset to the old one.



## Where it lives

| <b>Document</b>  | xSection (f8deedeb1fbd819a8fa20113), 'Stiffness Tools'                                                     |
|------------------|------------------------------------------------------------------------------------------------------------|
| <b>Tab</b>       | updateProfile (+ tables/profileUpdatesTable.fs)                                                            |
| <b>Tests</b>     | none                                                                                                       |
| <b>Examples</b>  | Test studio: Update profile 1                                                                              |
| <b>Explainer</b> | docs/explainers/xsection/xsection_explained.md, section 2.7                                                |
| <b>Related</b>   | El and Cross Section, Estimate Stiffness, Estimate Deflection, Generate / Analyze baseline, Update profile |